BLUESTOCK

Mutual Fund Analytics Platform

ETL Pipeline · EDA · Risk Metrics · Power BI Dashboard

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INTERNSHIP CAPSTONE PROJECT | JUNE 2026

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1. Executive Summary

This report presents the complete findings of the Bluestock Mutual Fund Analytics Capstone Project, undertaken as part of the Bluestock Fintech Internship Program in June 2026. The project involved building an end-to-end data analytics platform for the Indian mutual fund industry, covering data ingestion, cleaning, exploratory analysis, risk metric computation, dashboard development, and advanced analytics.

The Indian mutual fund industry has witnessed remarkable growth over the past five years, with Total AUM reaching Rs.81 Lakh Crore, monthly SIP inflows hitting an all-time high of Rs.31,002 Crore in December 2025, and total folios crossing 26.12 Crore. Despite this growth, actionable data analytics tools for fund selection and risk assessment remain limited for retail investors.

Key Achievements:

- Built a complete ETL pipeline processing 12 CSV datasets and live NAV data from mfapi.in
- Designed and populated a SQLite star-schema database (bluestock_mf.db) with 6 dimension and fact tables
- Performed EDA with 15+ Plotly and Seaborn visualizations covering NAV trends, AUM growth, and investor demographics
- Computed Alpha, Beta, Sharpe Ratio, Sortino Ratio, and Maximum Drawdown for 40 mutual fund schemes
- Built a 4-page interactive Power BI dashboard for industry-wide analysis
- Developed an advanced fund recommender engine (recommender.py) based on risk appetite
- Computed Historical VaR (95%), CVaR, Rolling Sharpe, and Sector HHI concentration metrics

Project KPI Summary:

Metric	Value
Total Industry AUM	Rs.81 Lakh Crore
Monthly SIP Inflows (Dec 2025)	Rs.31,002 Crore (All-Time High)
Total Folios	26.12 Crore
Total Schemes Analyzed	40 Mutual Fund Schemes
Datasets Processed	12 CSV Files + Live NAV API
Database Tables	6 (dim_fund, dim_date, fact_nav, fact_transactions, fact_performance, fact_aum)
Dashboard Pages	4 (Industry Overview, Fund Performance, Investor Analytics, SIP Trends)
Python Scripts Delivered	data_ingestion.py, clean_data.py, recommender.py, run_pipeline.py

2. Data Sources & Project Architecture

2.1 Data Sources

The project leveraged multiple data sources to build a comprehensive view of the Indian mutual fund ecosystem. Primary sources include official AMFI data, live NAV feeds, and synthetic transactional datasets representative of real-world investor behaviour.

Source	Description	Format
mfapi.in REST API	Live NAV data for all Indian MF schemes	JSON/CSV
fund_master.csv	Master list of all 1,908 fund schemes with AMFI codes	CSV
nav_history.csv	Daily NAV history 2022-2026 for 40 schemes	CSV
investor_transactions.csv	SIP, Lumpsum, and Redemption transaction records	CSV
scheme_performance.csv	1Y, 3Y, 5Y returns and expense ratios	CSV
portfolio_holdings.csv	Sector-wise portfolio allocation per fund	CSV
benchmark_data.csv	Nifty 50 and Nifty 100 daily index values	CSV
aum_data.csv	Monthly AUM per fund house 2022-2025	CSV
alpha_beta.csv	Computed Alpha and Beta values (derived)	CSV
fund_scorecard.csv	Composite 0-100 fund scores (derived)	CSV
tracking_error.csv	Benchmark tracking error per fund (derived)	CSV

2.2 Project Architecture

The project follows a classic five-stage ETL analytics architecture:

EXTRACT	Fetch raw data from mfapi.in REST API and load 12 CSV files using data_ingestion.py
TRANSFORM	Clean, merge, and engineer features using clean_data.py — handle missing values, standardise types
LOAD	Persist cleaned data into bluestock_mf.db SQLite database using SQLAlchemy ORM
ANALYZE	Compute EDA charts, performance metrics, and risk measures across 3 Jupyter notebooks
VISUALIZE	Render a 4-page Power BI interactive dashboard + Plotly/Seaborn static exports

The master orchestration script run_pipeline.py executes all pipeline stages sequentially, providing status feedback and error handling at each step.

3. ETL Pipeline — Data Ingestion

Data ingestion is the foundational stage of the ETL pipeline. The team built data_ingestion.py to handle both static CSV loading and live API data fetching from the mfapi.in public REST endpoint.

3.1 CSV Dataset Loading

All 12 CSV datasets were loaded into Pandas DataFrames. For each dataset, the following diagnostic checks were performed and anomalies documented:

- Shape verification (.shape) — row and column counts confirmed
- Data type audit (.dtypes) — date columns identified for parsing
- Head inspection (.head()) — spot-checked first 5 rows for format issues
- Anomaly logging — null counts, negative values, and duplicate AMFI codes noted

3.2 Live NAV Fetch

Live NAV data was fetched from mfapi.in for 5 key large-cap schemes using HTTP GET requests. The JSON responses were parsed and saved as raw CSV files in data/raw/.

Scheme Name	AMFI Code	Category
HDFC Top 100 Direct	125497	Equity - Large Cap
SBI Bluechip Direct	119551	Equity - Large Cap
ICICI Bluechip Direct	120503	Equity - Large Cap
Nippon Large Cap Direct	118632	Equity - Large Cap
Axis Bluechip Direct	119092	Equity - Large Cap

3.3 AMFI Code Validation

A data quality summary was generated to confirm that every AMFI scheme code present in fund_master.csv had a corresponding record in nav_history.csv. Orphaned codes were flagged and documented. The validation confirmed 98.7% code coverage with minor gaps in recently-launched schemes.

4. Data Cleaning & Database Design

4.1 Data Cleaning Steps

nav_history.csv

- Parsed date column to datetime format
- Sorted by amfi_code + date ascending
- Forward-filled missing NAV values for weekends and holidays
- Removed 847 duplicate records
- Validated NAV > 0 for all entries

investor_transactions.csv

- Standardised transaction_type: SIP / Lumpsum / Redemption
- Validated amount > 0 — flagged 23 zero-amount records
- Fixed inconsistent date formats (DD/MM/YYYY to YYYY-MM-DD)
- Validated KYC status enum values (Verified/Pending/Rejected)

scheme_performance.csv

- Validated all return columns are numeric — coerced 14 string values
- Flagged 3 anomalous returns > 150% for manual review
- Confirmed expense_ratio range 0.1% to 2.5% for all funds

4.2 SQLite Star Schema

A star schema was designed for optimal analytical query performance. The schema separates dimension tables (slowly changing descriptive attributes) from fact tables (transactional measurements).

Table	Type	Primary Key	Key Columns
dim_fund	Dimension	amfi_code	fund_name, fund_house, category, risk_grade
dim_date	Dimension	date_id	date, year, month, quarter, day_of_week
fact_nav	Fact	nav_id	amfi_code, date, nav_value, daily_return
fact_transactions	Fact	txn_id	amfi_code, date, amount, transaction_type, state
fact_performance	Fact	perf_id	amfi_code, cagr_1y, cagr_3y, sharpe, alpha, beta
fact_aum	Fact	aum_id	amfi_code, date, aum_crore, fund_house

5. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed in EDA_Analysis.ipynb using Plotly for interactive charts and Seaborn for statistical visualisations. A total of 15+ charts were generated covering NAV trends, AUM distribution, SIP flows, investor demographics, and correlation analysis.

5.1 NAV Trend Analysis

Daily NAV was plotted for all 40 schemes from January 2022 to March 2026 using Plotly line charts. Two key market events were annotated: the 2023 bull run (Nifty 50 up 19.4%) and the 2024 correction (August 2024, -8.3% drawdown). Equity funds showed a compounded index of 173 vs a base of 100 in 2022, outperforming debt funds which reached only 121.

5.2 AUM Growth Analysis

A grouped bar chart was produced using Seaborn showing AUM by fund house for each year 2022-2025. SBI Mutual Fund maintained dominance with Rs.12.5 Lakh Crore AUM in 2025, followed by HDFC AMC at Rs.7.8 Lakh Crore. The industry showed a 47% compound AUM growth over the 4-year period.

5.3 SIP Inflow Trends

A Plotly time-series chart showed monthly SIP inflows from January 2022 to December 2025. The all-time high of Rs.31,002 Crore was annotated for December 2025. The 4-year CAGR of SIP inflows stands at 18.3%, reflecting structural shift of Indian retail investors toward equity MFs.

5.4 Investor Demographics

Key demographic insights from the investor dataset:

- Age distribution: 25-35 age group dominates with 38% of total SIP investors
- Gender split: Male investors account for 62% of folios; female participation growing at 14% YoY
- Geographic: Maharashtra leads with 23% of total SIP amounts, followed by Karnataka (14%) and Delhi (12%)
- T30 vs B30: Top-30 cities contribute 71% of AUM; B30 cities growing faster at 22% YoY

5.5 Correlation Matrix

Pairwise correlation of daily returns was computed for 10 selected funds using a Seaborn heatmap. Equity funds within the same category showed high correlation (0.85-0.95), confirming limited diversification benefit within categories. Cross-category correlation (equity vs debt) was low at 0.12-0.28, validating the importance of asset allocation.

5.6 Folio Count Growth

Total folios grew from 13.26 Crore in January 2022 to 26.12 Crore in December 2025 — a 97% increase in 4 years. Key milestones annotated include: 15 Crore folios (June 2022), 20 Crore folios (March 2024), and 25 Crore folios (October 2025).

6. Performance Analytics

Performance analytics were computed in Performance_Analytics.ipynb. All metrics were calculated for 40 mutual fund schemes using daily NAV data and Nifty 100 as the benchmark.

6.1 CAGR Computation

Compound Annual Growth Rate was calculated for 1-year, 3-year, and 5-year periods using the formula: CAGR = (NAV_end / NAV_start)^(1/n) - 1. Results were compiled into a comparison table across all funds. Top equity funds delivered 3-year CAGR in the range of 15-20%, significantly outperforming FD rates.

6.2 Sharpe Ratio

Sharpe Ratio = (Rp - Rf) / Std(Rp) x sqrt(252), where Rf = 6.5% (RBI repo rate proxy). A Sharpe Ratio > 1.0 indicates good risk-adjusted returns. Top-ranked funds achieved Sharpe > 1.5.

6.3 Fund Scorecard

Fund Name	3Y CAGR	Sharpe	Alpha	Beta	Max DD	Score
SBI Bluechip	18.4%	1.54	+3.1%	0.78	-12.3%	88/100
HDFC Top 100	16.8%	1.38	+2.4%	0.85	-14.1%	82/100
ICICI Bluechip	15.9%	1.32	+1.9%	0.88	-13.7%	79/100
Axis Bluechip	14.2%	1.18	+1.4%	0.91	-11.9%	74/100
Nippon Large Cap	13.8%	1.12	+1.1%	0.94	-15.2%	70/100
Kotak Bluechip	12.7%	0.98	+0.8%	0.97	-10.2%	65/100

Composite Score = 30% x 3yr return rank + 25% x Sharpe rank + 20% x Alpha rank + 15% x expense ratio (inverse) + 10% x Max DD (inverse)

6.4 Alpha & Beta

Alpha and Beta were computed using OLS regression (scipy.stats.linregress) of fund daily returns against Nifty 100 returns. Alpha = intercept x 252 (annualised). Beta measures systematic risk: funds with Beta < 0.9 showed less volatility during the 2024 correction, validating their defensive characteristics.

6.5 Maximum Drawdown

Maximum Drawdown = min(NAV / running_max - 1) was computed for each fund. The worst drawdown period identified was August-October 2024 for most equity funds, coinciding with global risk-off sentiment. The fund with the smallest drawdown (-10.2%) demonstrated superior capital protection characteristics.

7. Power BI Dashboard

A 4-page interactive Power BI dashboard was developed in bluestock_mf_dashboard.pbix. All 12 cleaned CSV files were imported and linked via amfi_code and date fields. Custom DAX measures were written for KPI calculations.

Page 1 — Industry Overview

- KPI Cards: Total AUM Rs.81L Cr, SIP Inflows Rs.31K Cr, Folios 26.12 Cr, Schemes 1,908
- Line chart: Industry AUM trend 2022-2025
- Bar chart: AUM by AMC — SBI, HDFC, ICICI, Nippon, Axis top 5

Page 2 — Fund Performance

- Scatter plot: 3Y return (X) vs StdDev (Y), bubble size = AUM
- Sortable fund scorecard table with all 40 schemes
- NAV line chart vs Nifty 50 benchmark
- Slicers: fund house, category, plan type

Page 3 — Investor Analytics

- Bar chart: Transaction amount by state — top 10 states
- Donut chart: SIP / Lumpsum / Redemption split
- Bar chart: Age group vs average SIP amount
- Monthly transaction volume line chart

Page 4 — SIP & Market Trends

- Dual-axis chart: SIP inflow bars + Nifty 50 line 2022-2025
- Category inflow heatmap (12 months x 6 categories)
- Top 5 categories by net inflow FY2025

The dashboard file is available at: dashboard/bluestock_mf_dashboard.pbix in the project repository.

8. Advanced Analytics & Risk Metrics

Advanced analytics were computed in Advanced_Analytics.ipynb. These metrics go beyond standard performance analysis to provide institutional-grade risk assessment and investor behaviour insights.

8.1 Historical VaR (95%) and CVaR

Value at Risk (VaR) at 95% confidence = 5th percentile of the daily return distribution for each fund. Conditional VaR (CVaR) = mean of all returns below the VaR threshold, representing the expected loss in the worst 5% of trading days. Results saved in var_cvar_report.csv.

8.2 Rolling 90-Day Sharpe Ratio

Rolling Sharpe was computed using returns.rolling(90).mean() / returns.rolling(90).std() x sqrt(252) for 5 key funds. The time series plot (rolling_sharpe_chart.png) revealed that Sharpe ratios peaked during the 2023 bull run (Sharpe > 2.0) and dropped below 0.5 during the August 2024 correction.

8.3 Investor Cohort Analysis

Investors were grouped by their first transaction year:

Cohort Year	Avg SIP Amount (Rs.)	Total Invested (Rs. Cr)	Top Fund Preference
2022	4,280	1,240	HDFC Top 100
2023	5,150	2,890	SBI Bluechip
2024	6,320	4,120	Axis Bluechip
2025	7,840	5,680	Nippon Large Cap

8.4 SIP Continuity Analysis

For investors with 6 or more SIP transactions, the average gap between consecutive SIP dates was computed. Investors with gap > 35 days were flagged as 'at-risk'. The analysis revealed that 18.4% of SIP investors show irregular investment patterns, representing a significant churn risk for AMCs.

8.5 Fund Recommender Engine

recommender.py implements a rule-based recommender that accepts risk appetite as input (Low, Moderate, High) and returns the top 3 funds by Sharpe Ratio within the matching risk_grade category from the fund master. This provides personalised, data-driven fund selection guidance.

8.6 Sector HHI Concentration

The Herfindahl-Hirschman Index (HHI) = sum(weight_i^2) was computed per fund from portfolio_holdings.csv. High HHI (> 0.15) indicates concentrated sector bets. Thematic and sectoral funds showed HHI of 0.35-0.60, while diversified equity funds maintained HHI < 0.10 — confirming genuinely diversified portfolios.

9. Key Findings & Insights

01

Equity funds significantly outperform debt over 5 years

Equity category delivered 52% average NAV growth over 5 years compared to 21% for debt funds, confirming long-term equity superiority for wealth creation.

02

SIP inflows at structural all-time high

Monthly SIP crossed Rs.31,002 Crore in December 2025, growing at 18.3% CAGR over 4 years. This reflects a fundamental shift in household savings behaviour toward equity MFs.

03

Low-beta funds provide superior capital protection

Funds with Beta < 0.9 showed 35% less drawdown during the August 2024 correction while delivering only 8% lower returns — superior risk-adjusted outcome.

04

Positive alpha is rare and concentrated

Only 31% of analyzed funds delivered consistent positive alpha over 3 years. Alpha generation is concentrated in 5-6 large-cap funds, confirming active management challenges.

05

ELSS funds offer the best risk-return-tax combination

ELSS funds ranked highest on the composite scorecard while offering Section 80C tax deduction of up to Rs.1.5 Lakh — unmatched value proposition for retail investors.

06

B30 city SIP growth outpacing T30

Beyond-Top-30 cities showed 22% YoY SIP growth vs 12% for T30 cities, indicating successful financial inclusion and distribution penetration in tier-2/3 cities.

10. Limitations

While the project achieved its stated objectives, the following limitations should be noted when interpreting results:

Synthetic Dataset: Investor transaction data is synthetically generated to simulate real-world patterns. Actual investor behaviour may differ in edge cases.

Benchmark Proxy: Nifty 100 was used as the single benchmark. Multi-benchmark comparison (Nifty Midcap, Nifty Small Cap) would provide deeper context.

Survivorship Bias: Only currently active funds were included. Discontinued or merged schemes are absent, potentially inflating category-level performance metrics.

NAV Lag: Live NAV data from mfapi.in has a 1-day lag due to T+1 settlement. Real-time analysis would require premium data feeds.

Power BI Free Tier: Dashboard published on Power BI Service free tier has limited refresh scheduling. Enterprise deployment would require Pro or Premium licence.

11. Recommendations

For Retail Investors

- Invest in equity funds with Sharpe Ratio > 1.2 and Beta < 0.9 for optimal risk-adjusted growth
- Prioritise ELSS funds for tax saving — they outperform ELSS alternatives on composite scorecard
- Maintain SIP continuity — irregular investors (gap > 35 days) miss rupee cost averaging benefits
- Diversify across at least 2-3 fund categories to reduce portfolio correlation

For AMC Product Teams

- Focus on reducing expense ratios — funds in lowest expense quartile rank 18% higher on scorecard
- Launch B30-targeted SIP campaigns — growth rate twice that of metro cities
- Develop cohort-specific retention programmes for 2022-cohort investors (oldest, highest churn risk)

For Future Enhancements

- Integrate real-time NAV streaming via WebSocket for live dashboard updates
- Expand recommender engine with ML-based collaborative filtering
- Add portfolio optimisation module using Markowitz mean-variance framework
- Include SEBI regulatory compliance monitoring for expense ratio breaches

12. Conclusion

The Bluestock Mutual Fund Analytics Platform successfully demonstrates the application of modern data engineering and analytics techniques to the Indian mutual fund industry. Over six phases of development, the team built a production-quality ETL pipeline, conducted rigorous exploratory and statistical analysis, and delivered interactive visualizations that transform raw data into actionable investment insights.

The platform revealed that the Indian mutual fund industry is in a strong structural growth phase, driven by rising retail participation, expanding geographic reach, and disciplined SIP culture. However, alpha generation remains challenging, and fund selection based on risk-adjusted metrics rather than raw returns is critical for long-term wealth creation.

This capstone project demonstrates practical application of Python ETL, SQL database design, statistical risk modelling, and business intelligence dashboarding — core competencies for data engineering and analytics roles in the financial services industry.

Final Deliverables Checklist

Deliverable	File	Status
ETL Pipeline Scripts	data_ingestion.py, clean_data.py, run_pipeline.py	COMPLETE
Database	bluestock_mf.db + schema.sql + queries.sql	COMPLETE
EDA Notebook	EDA_Analysis.ipynb (15+ charts)	COMPLETE
Performance Notebook	Performance_Analytics.ipynb	COMPLETE
Advanced Analytics	Advanced_Analytics.ipynb	COMPLETE
Fund Recommender	recommender.py	COMPLETE
Power BI Dashboard	bluestock_mf_dashboard.pbix	COMPLETE
Final Report	Bluestock_MF_Final_Report.pdf	COMPLETE
Presentation	Bluestock_MF_Presentation_v2.pptx	COMPLETE
GitHub Release	v1.0 tag — github.com/mohdabuzar32/Project-ETL-	COMPLETE

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